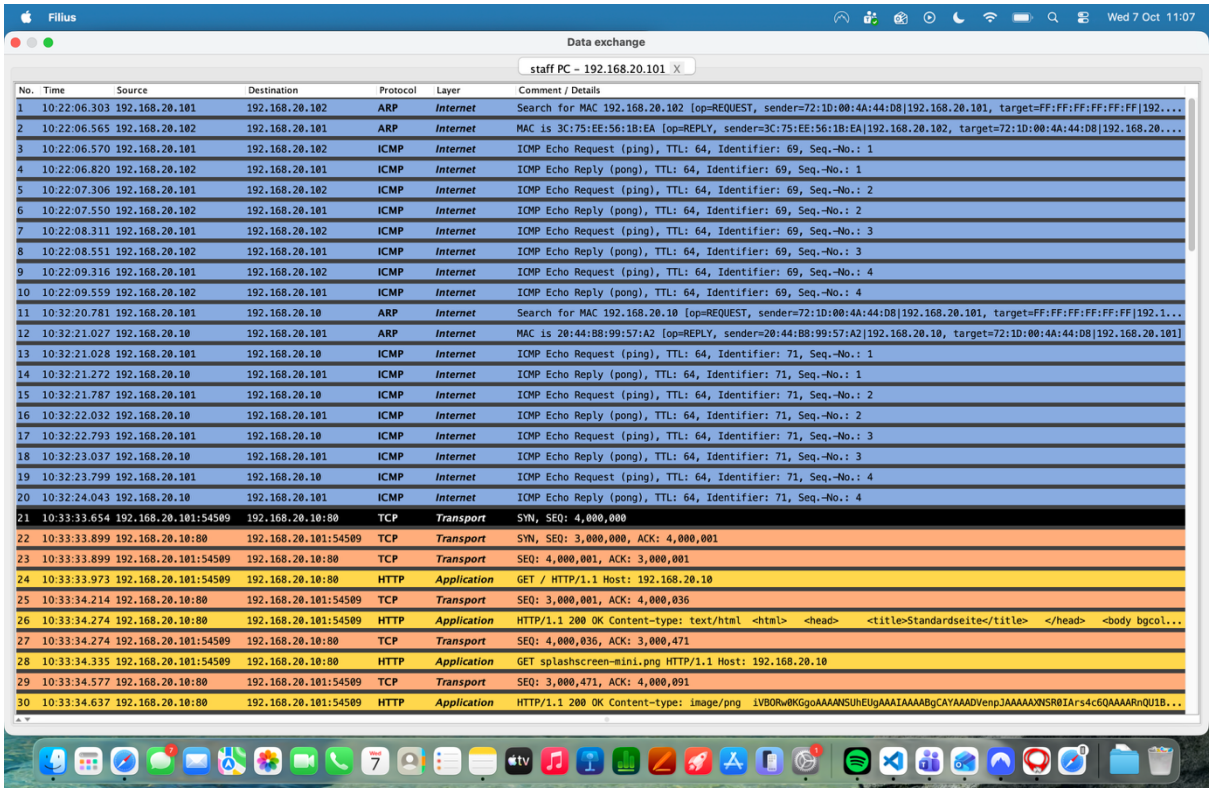
Screenshot 1 – Protocol Data Exchange

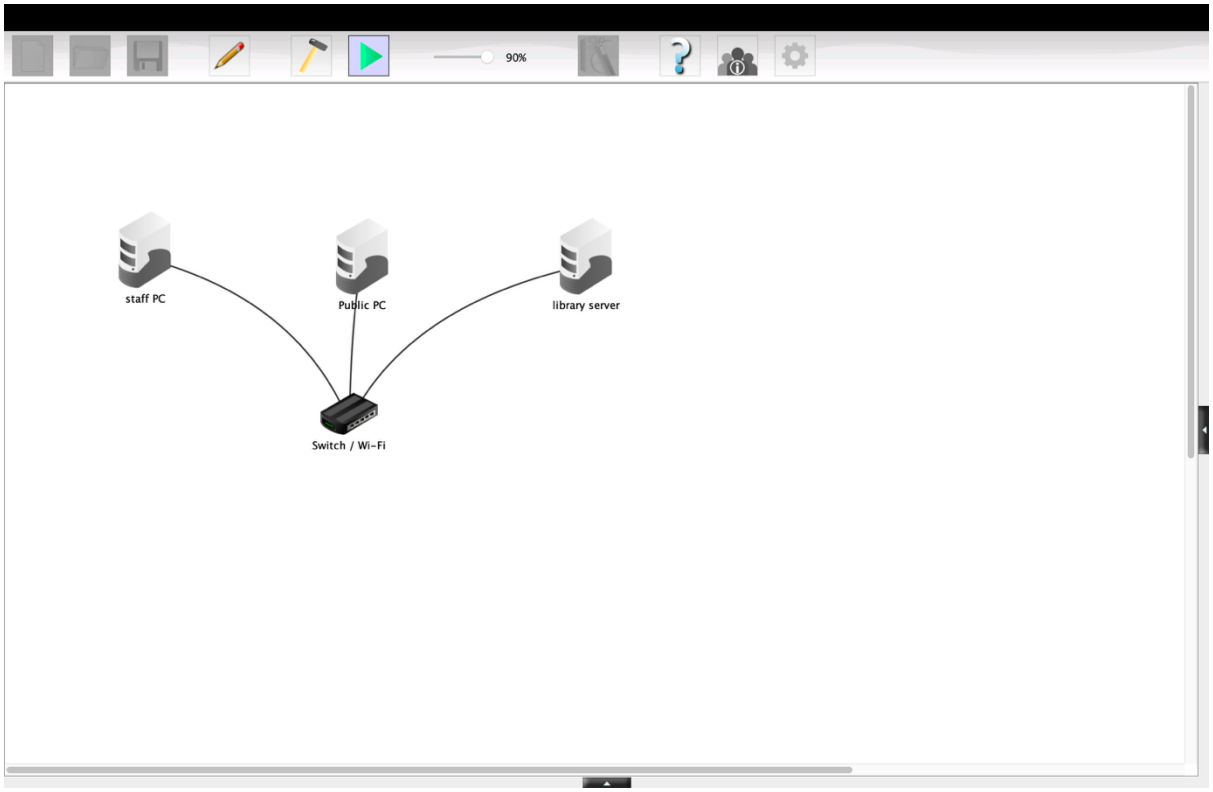
This screenshot shows the Data Exchange window in Filius. ARP was used to find the MAC address of devices on the network, while ICMP was used for the ping requests and replies. TCP was used to establish a reliable connection with the Library Server, and HTTP was used to request and receive the web page.



No.	Time	Source	Destination	Protocol	Layer	Comment / Details
1	10:22:06.303	192.168.20.101	192.168.20.102	ARP	Internet	Search for MAC 192.168.20.102 [op=REQUEST, sender=72:1D:00:4A:44:D8|192.168.20.101, target=FF:FF:FF:FF:FF:FF|192....
2	10:22:06.565	192.168.20.102	192.168.20.101	ARP	Internet	MAC is 3C:75:EE:56:1B:EA [op=REPLY, sender=3C:75:EE:56:1B:EA|192.168.20.102, target=72:1D:00:4A:44:D8|192.168.20....
3	10:22:06.570	192.168.20.101	192.168.20.102	ICMP	Internet	ICMP Echo Request (ping), TTL: 64, Identifier: 69, Seq.-No.: 1
4	10:22:06.820	192.168.20.102	192.168.20.101	ICMP	Internet	ICMP Echo Reply (pong), TTL: 64, Identifier: 69, Seq.-No.: 1
5	10:22:07.306	192.168.20.101	192.168.20.102	ICMP	Internet	ICMP Echo Request (ping), TTL: 64, Identifier: 69, Seq.-No.: 2
6	10:22:07.550	192.168.20.102	192.168.20.101	ICMP	Internet	ICMP Echo Reply (pong), TTL: 64, Identifier: 69, Seq.-No.: 2
7	10:22:08.311	192.168.20.101	192.168.20.102	ICMP	Internet	ICMP Echo Request (ping), TTL: 64, Identifier: 69, Seq.-No.: 3
8	10:22:08.551	192.168.20.102	192.168.20.101	ICMP	Internet	ICMP Echo Reply (pong), TTL: 64, Identifier: 69, Seq.-No.: 3
9	10:22:09.316	192.168.20.101	192.168.20.102	ICMP	Internet	ICMP Echo Request (ping), TTL: 64, Identifier: 69, Seq.-No.: 4
10	10:22:09.559	192.168.20.102	192.168.20.101	ICMP	Internet	ICMP Echo Reply (pong), TTL: 64, Identifier: 69, Seq.-No.: 4
11	10:32:20.781	192.168.20.101	192.168.20.10	ARP	Internet	Search for MAC 192.168.20.10 [op=REQUEST, sender=72:1D:00:4A:44:D8|192.168.20.101, target=FF:FF:FF:FF:FF:FF|192.1...
12	10:32:21.027	192.168.20.10	192.168.20.101	ARP	Internet	MAC is 20:44:B8:99:57:A2 [op=REPLY, sender=20:44:B8:99:57:A2|192.168.20.10, target=72:1D:00:4A:44:D8|192.168.20.101]
13	10:32:21.028	192.168.20.101	192.168.20.10	ICMP	Internet	ICMP Echo Request (ping), TTL: 64, Identifier: 71, Seq.-No.: 1
14	10:32:21.272	192.168.20.10	192.168.20.101	ICMP	Internet	ICMP Echo Reply (pong), TTL: 64, Identifier: 71, Seq.-No.: 1
15	10:32:21.787	192.168.20.101	192.168.20.10	ICMP	Internet	ICMP Echo Request (ping), TTL: 64, Identifier: 71, Seq.-No.: 2
16	10:32:22.032	192.168.20.10	192.168.20.101	ICMP	Internet	ICMP Echo Reply (pong), TTL: 64, Identifier: 71, Seq.-No.: 2
17	10:32:22.793	192.168.20.101	192.168.20.10	ICMP	Internet	ICMP Echo Request (ping), TTL: 64, Identifier: 71, Seq.-No.: 3
18	10:32:23.037	192.168.20.10	192.168.20.101	ICMP	Internet	ICMP Echo Reply (pong), TTL: 64, Identifier: 71, Seq.-No.: 3
19	10:32:23.799	192.168.20.101	192.168.20.10	ICMP	Internet	ICMP Echo Request (ping), TTL: 64, Identifier: 71, Seq.-No.: 4
20	10:32:24.043	192.168.20.10	192.168.20.101	ICMP	Internet	ICMP Echo Reply (pong), TTL: 64, Identifier: 71, Seq.-No.: 4
21	10:33:33.654	192.168.20.101:54509	192.168.20.10:80	TCP	Transport	SYN, SEQ: 4,000,000
22	10:33:33.899	192.168.20.10:80	192.168.20.101:54509	TCP	Transport	SYN, SEQ: 3,000,000, ACK: 4,000,001
23	10:33:33.899	192.168.20.101:54509	192.168.20.10:80	TCP	Transport	SEQ: 4,000,001, ACK: 3,000,001
24	10:33:33.973	192.168.20.101:54509	192.168.20.10:80	HTTP	Application	GET / HTTP/1.1 Host: 192.168.20.10
25	10:33:34.214	192.168.20.10:80	192.168.20.101:54509	TCP	Transport	SEQ: 3,000,001, ACK: 4,000,036
26	10:33:34.274	192.168.20.10:80	192.168.20.101:54509	HTTP	Application	HTTP/1.1 200 OK Content-type: text/html <html> <head> <title>Standardseite</title> </head> <body bgcol...
27	10:33:34.274	192.168.20.101:54509	192.168.20.10:80	TCP	Transport	SEQ: 4,000,036, ACK: 3,000,471
28	10:33:34.335	192.168.20.101:54509	192.168.20.10:80	HTTP	Application	GET splashscreen-mini.png HTTP/1.1 Host: 192.168.20.10
29	10:33:34.577	192.168.20.10:80	192.168.20.101:54509	TCP	Transport	SEQ: 3,000,471, ACK: 4,000,091
30	10:33:34.637	192.168.20.10:80	192.168.20.101:54509	HTTP	Application	HTTP/1.1 200 OK Content-type: image/png iVBORw0KGgoAAAANSUhEUgAAAIAAAABgCAYAAADVenpJAAAAAXNSR0IArs4c6QAAAARnQU1B...

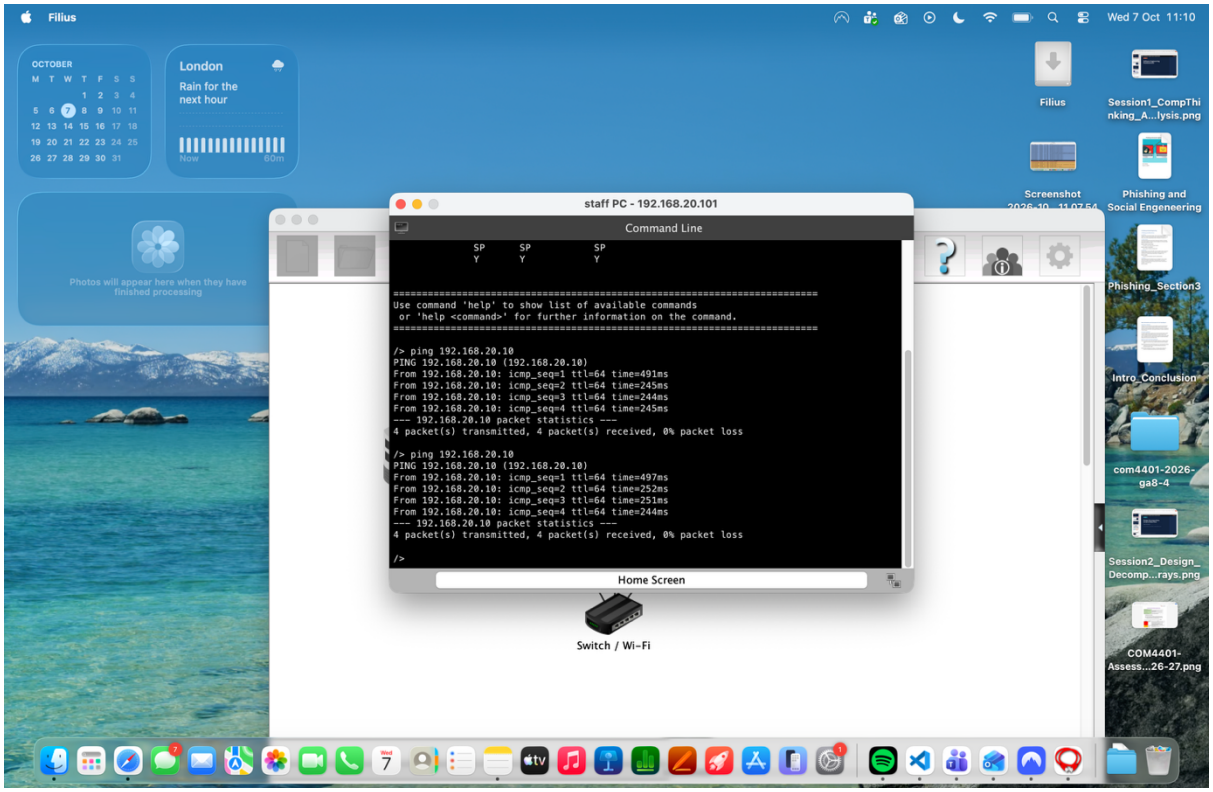
Screenshot 2 – Network Topology

This screenshot shows the LAN created in Filius. The Staff PC, Public PC and Library Server are connected to the same switch, allowing the devices to communicate with each other on the local network.



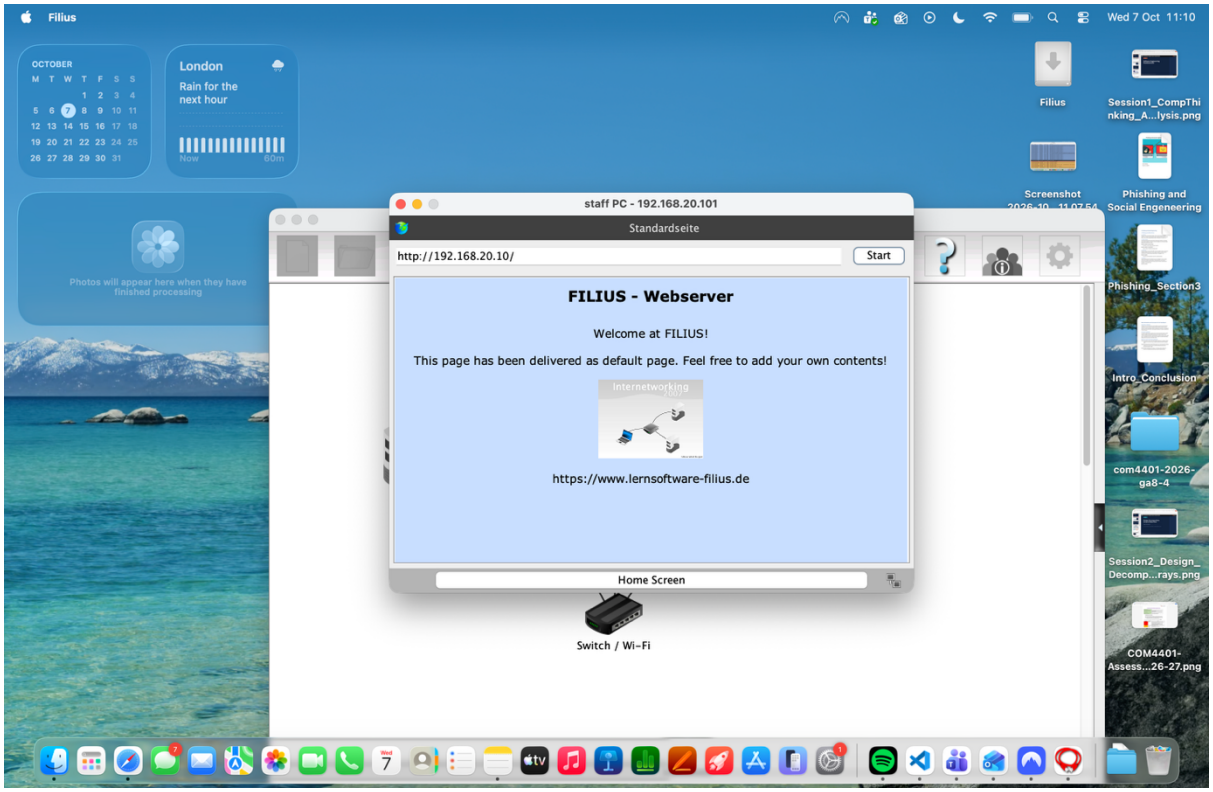
Screenshot 3 – Successful Ping Test

This screenshot shows a successful ping from the Staff PC (192.168.20.101) to the Library Server (192.168.20.10). All four packets were received with 0% packet loss, confirming that the Staff PC can successfully communicate with the server.



Screenshot 4 – Web Server Test

This screenshot shows the Staff PC successfully accessing a web page hosted by the Library Server using the address http://192.168.20.10. This demonstrates that the client can request and receive a web service from the server using HTTP.



The Staff PC acts as the client, and the Library Server acts as the server. The successful ping confirms that the devices can communicate across the LAN. When the Staff PC requests the web page, the Library Server responds and sends the page back. This demonstrates a basic client-server relationship.

ARP
Top layer reached: Network
What it did: Finds the MAC address linked to an IP address.

ICMP
Top layer reached: Internet
What it did: Sends ping request and reply to messages to check if another device can be reached.

TCP
Top layer reached: Transport
What it did: Creates a reliable connection between the client and the server.

HTTP
Top layer reached: Application
What it did: Requests and transfers the web page between the client and the server.

DNS (Extension)
Top layer reached: Application
What it did: Converts a domain name into an IP address.

Why did ping use fewer layers than loading the web page?

Ping uses fewer layers because it only checks whether another device can be reached, so it uses ICMP at the Internet layer. Loading a web page needs a reliable TCP connection and an HTTP request at the Application layer. This means loading a web page uses more layers because it requires more communication between the client and server.